

B T G V PRASAD CHATRATHI

Computer Vision Engineer | Machine Learning Engineer | AI Engineer

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PROFESSIONAL SUMMARY

Computer Vision / Machine Learning Engineer with 2 years of experience building and deploying real-time computer vision systems for manufacturing and industrial applications. Experienced in the end-to-end ML lifecycle including data collection, dataset curation, annotation, training, evaluation, optimization, deployment, and production monitoring. Skilled in YOLO, PyTorch, CNNs, OpenCV, object detection, classification, segmentation, tracking, and video analytics. Experienced in analyzing real-world model failures, identifying hard samples, improving datasets through targeted collection and augmentation, and optimizing models for accuracy, latency, and real-time performance. Hands-on experience developing production ML systems, REST APIs, and containerized deployments using Python, FastAPI, Flask, and Docker, with a focus on continuous improvement based on field performance.

PROFESSIONAL EXPERIENCE

Computer Vision Engineer

ImageVision.ai

Nov 2024 - Present

Hyderabad, India

- Built end-to-end computer vision pipelines covering data collection, dataset curation, annotation, model training, evaluation, optimization, deployment, and production debugging for industrial applications.
- Developed and optimized YOLOv5/YOLOv8 and CNN models using PyTorch, data augmentation, class balancing, and hyperparameter tuning to improve precision, recall, and real-world robustness.
- Analyzed real-world model failures and hard samples involving occlusion, variable lighting, camera-angle variation, small objects, and false detections; improved datasets through targeted data collection and augmentation.
- Implemented real-time video analytics, object tracking, and temporal event analysis, and deployed production inference services using Python, OpenCV, FastAPI, Flask, and Docker; supported field monitoring and continuous model improvement.

Facial Recognition System - Real-Time Biometric AI System

- Business Problem: Manual identity verification (e.g., attendance, ID checks) is slow and error-prone.
- Solution: Built a real-time facial recognition pipeline using DeepFace with FaceNet-512 embeddings, integrated with iris authentication for a unified biometric verification system.
- Results: Achieved 90%+ identification accuracy across variable lighting, partial occlusion, and diverse demographics; validated as a proof-of-concept for college attendance and government ID verification use cases.

Technologies: Python, OpenCV, Dlib, YOLOv5, CNN, DeepFace, FaceNet-512, PyTorch, Transfer Learning

Automated Box Counting on Conveyor Lines

- Business Problem: Manual box counting on conveyor lines is slow, error-prone, and doesn't scale with production throughput, leading to inventory mismatches and delayed dispatch reporting.
- Solution: Real-time object detection and counting on existing CCTV/camera infrastructure, requiring no additional hardware, with automated logging and reporting of box counts per shift/line.
- Results: Achieved 98% counting accuracy in production, eliminating manual tally errors and enabling real-time throughput visibility.

Technologies: Python, YOLOv8, OpenCV, PyTorch, Object Detection, Tracking Algorithms, Real-Time Inference

360° Glass Bottle Inspection System

- Business Problem: Manual visual inspection of glass bottles cannot reliably detect micro-level defects (as small as 0.5mm) across the full 360° surface, leading to inconsistent quality control and defective bottles reaching packaging/dispatch.
- Solution: Automated 360° glass-in-glass inspection system using computer vision to detect surface and structural defects including black particles, mold contamination, blisters, and cut marks down to 0.5mm in size, integrated into the bottling line without additional manual checkpoints.
- Results: Achieved 90% detection accuracy in demo/pilot testing, validating feasibility for full production deployment.

Technologies: Python, YOLOv8, OpenCV, PyTorch, Object Detection, Defect Classification, Real-Time Inference

PPE Kit Detection & Worker Safety Monitoring - Security & Surveillance AI System

- Business Problem: Manual monitoring of PPE compliance (helmets, vests, gloves, masks, etc.) across large industrial/construction sites is inconsistent and cannot scale, increasing the risk of workplace accidents and regulatory non-compliance.
- Solution: Real-time detection of PPE kit compliance and worker safety violations on existing CCTV/surveillance infrastructure, requiring no additional hardware, with automated alerting for non-compliance (missing helmet, vest, gloves, etc.) and unsafe zone entry.
- Results: Improved PPE compliance monitoring coverage across the site, enabling faster violation detection and reducing manual surveillance workload.

Technologies: Python, YOLOv8, OpenCV, PyTorch, Object Detection, Tracking Algorithms, Real-Time Inference

TECHNICAL SKILLS

Programming Languages: Python

Computer Vision: OpenCV, YOLOv5, YOLOv8, Object Detection, Small Object Detection, Image Classification, Image Segmentation, Facial Recognition, Video Analytics

Machine Learning: Linear Regression, SVM, K-Means Clustering, Neural Networks

Deep Learning: CNN, Transfer Learning, Hyperparameter Tuning, Model Optimization, Real-Time Inference

Deployment: Flask, FastAPI, Streamlit, Docker, REST API

NLP / LLM: LangChain, FAISS, Vector Embeddings, Text Processing, Large Language Models

Data & Visualization: NumPy, Pandas, Tableau, Matplotlib

Soft Skills: Analytical Problem Solving, Critical Thinking, Cross-Functional Collaboration

EDUCATION

Bachelor of Technology (B. Tech), Electronics and Communication Engineering Feb 2020 - Apr 2024

Swarnandhra College of Engineering and Technology (Autonomous), Narasapuram, India | GPA: 7.58/10

CERTIFICATIONS

- Natural Language Processing on Google Cloud - Skills: NLP, Text Preprocessing, Vector Embeddings
 - Introduction to Computer Vision and Image Processing (IBM) - Skills: OpenCV, Image Processing, Object Detection, Image Segmentation, Pose Estimation
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VOLUNTEERING

- Team Lead, National Service Scheme, Narasapuram - Aug 2023